

Atmospheric Mediation

Addressing socio-spatial frictions in shared student housing through ambient interaction

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Abstract

Contemporary urban life is increasingly shaped by high-density, making shared domestic living a growing condition, particularly among students in transitional housing arrangements. Within these environments, everyday coexistence depends on subtle forms of negotiation around presence and availability. When these become ambiguous, socio-spatial frictions emerge through avoidance and waiting. While smart home technologies increasingly mediate domestic life, many existing systems prioritize automation and efficiency over the social dynamics of cohabitation. This thesis investigates how atmospheric interaction design can support cohabitation through ambient forms of communication embedded within the environment. Through a design process combining contextual inquiry, bodystorming, iterative prototyping, Wizard of Oz evaluations, and user enactments, the project developed an ambient lighting system that communicates occupancy and intention through low-resolution cues. The findings suggest that ambiguous and qualitative forms of communication can support coordination while preserving privacy, agency, and interpretative flexibility.

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1. INTRODUCTION

Contemporary urban life is increasingly shaped by high-density living, where shared households have become a common reality due to rising housing costs. In these contexts, co-living means managing different routines, habits, and expectations. Areas such as kitchens and bathrooms are viewed as spaces of overlap; daily movement and activities require ongoing regulation of presence and attention to others.

The student community represents a relevant portion of the affected group - many of whom are living away from home - often in temporary or transitional arrangements. For displaced students or those moving between phases of life, shared housing rarely means an ideal or long-term environment but rather a provisional solution determined by economic and social factors. Coexistence norms become undefined and inconsistent, requiring inhabitants to adapt to how space is used and shared.

Moments of discomfort are frequent when living together and coordination does not always happen through explicit communication. Instead, it relies on interpretation: reading the situation, anticipating others and adapting behavior accordingly. Many of these adjustments are subtle and often go unnoticed, yet they structure how common areas are used and experienced: deciding when to enter a determined room, whether to interrupt someone, or how long to occupy a space become social decisions instead of purely functional ones.

This reliance introduces a layer of uncertainty. A location may be occupied, but for how long? Someone may be present, but are they available and willing to interact? Over time, these small moments accumulate, shaping how people move in the interior. These situations often emerge from a lack of clear cues rather than resulting from a lack of space. The environment provides limited information about how it is being used, or how it is likely to be used next. Mediation therefore becomes an ongoing effort, requiring attention even in simple, everyday situations.

In addition, home technologies have become more oriented to support life at home. However, many of these systems focus on control, efficiency, or automation. In doing so, they assume a certain level of stability and established patterns of use that student accommodation frequently lacks. By requiring interaction through screens or devices, these technologies prioritize task-based management over the highly social processes through which people negotiate space. While these systems are effective for individual domestic tasks, they offer limited support to deal with the dynamic nature of the living environments.

This opens up a different line of inquiry: it becomes relevant to ask how the atmosphere itself might support interaction. If everyday coexistence relies on interpretation, could some of this process be made more perceptible? Could the environment provide signals that help people understand what is happening around them without requiring direct communication?

Existing work in ambient interaction has shown that information can be communicated through subtle changes in the surroundings, allowing users to maintain consciousness without shifting their attention. Even though ambient displays are being implemented for communicating various types of abstract data, their potential to deal with personal relations in residential settings remains yet to explore.

This thesis explores the potential of indicators to assist in the negotiation of space. The main goal is to reduce the energy this negotiation requires, not to eliminate ambiguity by implementing explicit systems or rules.

The project investigates the potential of small context-specific interventions in changing how people perceive and make use of the space, integrating interaction within the context to facilitate this coordination. By framing the environment as an active part of daily co-existence, this work seeks to enhance awareness and availability without explicit verbal communication.

The research question guiding this work is: *How can atmospheric interaction design mitigate social frictions in shared domestic environments?*

2. THEORETICAL FRAMEWORK AND RELATED WORK

2.1 Shared domestic environments as social and temporal systems

The concepts introduced in the previous section point toward a need to see and understand residential contexts as social and temporal systems shaped by routines and implicit ways of synchronization. While the increasing densification of urban environments has intensified the number of shared living arrangements (UN-Habitat, 2020), the challenges that emerge within these contexts are the result of how places are inhabited, shared, and redefined through everyday use.

The temporal organization of daily life lies at the core of this condition. According to Zerubavel (1981), time structuring is performed by socially understood yet personally implemented rhythms. Such rhythms are responsible for the timing and manner of performing certain actions, but are not necessarily synchronized between individuals. Different time structures coexist in one household, which results in overlapping of activities in incompatible ways. This idea is conceptualized through Lefebvre's (2004)

rhythmanalysis, which suggests that space is generated by the interaction of rhythms, and that mismatches happen when these rhythms collide. In this way, conflict within shared homes become inevitable as it is an inherent part of cohabitation.

These collisions are often subtle and found in everyday situations. These include times when the access to an area seems clear but is in fact confusing due to social implications, such as when a room is occupied but its usability is unknown, when presence does not clarify willingness to engage, or when multiple inhabitants want to use the same common location. These moments require interpretation of the environment from the individuals and for them to adjust their behavior accordingly, reinforcing the thesis main concern that coordination in shared living depends mainly on perception.

Furthermore, technological approaches to domestic settings have been prioritizing control over harmonization in the past years. Smart home systems have been designed as infrastructures for managing devices, focusing on efficiency, automation, and optimization (Owens et al., 2024). Interaction within these systems usually happens through interfaces like screens, requiring users to monitor and regulate their atmosphere.

2.2 Smart homes and control-oriented systems

This has been critiqued within Human-Computer Interaction: according to Crabtree et al. (2004), the home should be understood as a socially organized environment, shaped by routines, relationships, and tacit practices. Following this perspective, the true success of home technology is found in its ability to meet and align with existing patterns of living. This way, systems that impose new forms of interaction risk disrupting these patterns, introducing additional friction rather than reducing it.

This critique is extended by Felber et al. (2024), raising concerns about the capacity of technologically enhanced contexts to function as “homes” in an experiential sense. Their work questions how systems designed around efficiency and automation may compromise feelings of comfort and connection, especially when they do not consider the relational and affective dimensions of cohabitation. These critiques together suggest that current smart home works are insufficient for addressing the complexity and social dimension involved in communal living.

2.3 Space vs Place

It is necessary to reconsider how space is understood and defined within Interaction Design to move beyond control-oriented models. Harrison and Dourish (1996) provide

a distinction between space and place, arguing that while space refers to the physical properties of an environment, place emerges through use and meaning-making. In this sense, it relies on interpretation, reading the situation, anticipating others and adapting behavior accordingly; this process transforms the physical space of the home into a social place (Harrison & Dourish, 1996). This transformation is neither stable nor uniform in student dwelling contexts. Instead, it is continuously negotiated, as different inhabitants assign different meanings and expectations to the same configuration.

Benyon (2014) develops this by defining space as an interactive environment whose shape is informed by human experience. From this perspective, it actively participates in the production of meaning by shaping how situations are perceived, interpreted, and navigated. This suggests that the way space is designed and understood can influence how interactions unfold, particularly in everyday situations where inhabitants must continuously coordinate their activities and expectations.

2.4 Psychological and embodied implications of shared living

The consequences of this constant negotiation in cohabitation are also psychological. Evans (2003) demonstrates how the built environment can influence mental well-being, suggesting that the presence of conflict in the use of certain areas can not only cause stress and discomfort, but also increase difficulty in coordinating those areas. In a domestic context, where boundaries are often fluid and activities overlap, the lack of mechanisms for coordination can intensify these effects. These overlapping uses of space are amplified in high-density environments, such as student housing, increasing the chance of stress emerging from the continuous regulation of shared settings. This reinforces the need for residential mechanisms that can support management and reduce the cognitive load associated with everyday mediation.

The studies above have contributed for increasing interest in new ways of inserting the interaction within the environment itself. These approaches explore how information can be communicated through perceptual cues that integrate daily life, without making use of any kind of interface. Weiser (1991) describes this as a shift toward calm technology, where systems work in the periphery of attention, supporting awareness without demanding focus. This perspective becomes especially relevant in contexts where attention is already fragmented by ongoing routines and parallel activities.

Another valuable framework for understanding how these interactions can be designed comes from embodied interaction. Dourish (2001) argues that interaction is deeply grounded in our bodies and the physical world, emerging through action, movement, and spatial engagement. This perspective opens up the situated and relational nature of

interaction, treating it as more than an abstract exchange between user and system. Embodied approaches then offer a more appropriate model for design in collective dwellings, where interaction is often implicit and embedded in routine activity.

Ishii and Ullmer (1997) extend this point of view with Tangible Interaction by exploring how physical objects can mediate interaction. They propose that digital information can be represented through tangible forms, allowing users to engage with it through direct manipulation and physical arrangement. Hornecker and Buur (2006) build on this by highlighting the role of physical space and social context in shaping tangible interaction. Their framework emphasizes how interaction emerges through embodied engagement with artifacts and surrounding social situations. This suggests that objects may influence how situations are perceived and navigated within shared environments.

2.5 Social visibility, privacy and coordination breakdowns

The visibility of social information plays a major role in supporting coordination. Erickson and Kellogg (2000) introduce the concept of social translucence to describe systems that make aspects of human activity visible while preserving individual autonomy. Such systems enable individuals to adjust their behavior in relation to others without requiring explicit communication by providing understanding of others' intentions. Increasing its perceptibility becomes a key design opportunity in domestic contexts, where this information remains ambiguous.

The regulation of privacy complicates this dynamic, as we can see through Altman (1975) where privacy is conceptualized as a continuous process of boundary navigation, in which individuals seek to balance interaction and withdrawal. This process becomes collective in coexistence, requiring individuals to bargain their needs in relation to others. When signs about availability are unclear, this adjustment becomes more difficult, often leading to avoidance, hesitation and misinterpretation.

These challenges may be described as forms of socio-spatial friction, emerging as a result of convergence of temporal patterns, situated constraints, and social interaction. Such moments are described as collision points within this research: situations where multiple inhabitants attempt to use the same room in ways that are not immediately compatible. Existing strategies for managing these situations rely heavily on explicit communication, which can become socially draining over time.

2.6 Towards ambient and environmental mediation

The perspectives outlined above suggest that the challenges of shared living are not only a matter of physical design, but of how information about usability, time, and

social interaction is made available to inhabitants. There is an opportunity to explore how cues can support this process - by embedding information in the atmosphere, these hints can provide indirect communication that aligns with existing patterns of behavior.

Within this thesis, atmospheric interaction refers to forms of interaction embedded within the surrounding environment that communicate social and contextual information through subtle changes. These interactions work through peripheral awareness, shaping how inhabitants interpret presence and availability within domestic settings without demanding focused attention.

This research investigates how light can work as a mediator of facilitation. Among the different ways available for ambient and peripheral interaction, light is particularly suited for mediating social coordination in cohabitation households, because it operates without demanding focal attention and can be perceived incidentally within everyday activity. This allows it to function as a non-intrusive layer of communication embedded in the environment in order to support attentiveness without interrupting ongoing behavior. This is particularly relevant in shared living contexts, where inhabitants often seek to avoid additional cognitive burden when thinking about potential interactions. Light can signal presence, availability, or activity states in a way that remains subtle and non-intrusive, enabling discretion, aligning with the principles of calm technology and peripheral awareness.

This thesis brings together perspectives from smart home research, spatial interaction design, and social computing. It responds to existing critiques of control-based systems by proposing an alternative model in which the primary role of the environment is to support the negotiation of space. Through this approach, the project seeks to extend existing work on ambient and embedded interaction into the domain of shared occupancy, where the challenges are fundamentally social.

2.7 Canonical design cases in ambient interaction

While the theoretical perspectives outlined above establish the conceptual foundations for atmospheric interaction, it is equally important to examine how these ideas have been previously materialized through design practice. The following canonical design cases provide important precedents for understanding how atmospheric computation and home sensing have been approached within interaction design. These projects function as critical reference points from which this thesis extracts, questions, adapts, and challenges specific interaction principles.

Building upon Ishii and Ullmer's (1997) tangible interaction paradigms introduced in section 2.4, the *Ambient Room* (Ishii et al., 1998) remains a landmark exploration of how computational information can be embedded into physical space (appendix 01). The installation demonstrated how simulated natural phenomena - such as projected water airflow, and diffuse light - could operate as peripheral displays that communicate contextual information without demanding focused attention. From this work, this thesis adopts a key interaction principle: the use of environmental light as a subtle way of communicating changes in activity and presence. The *Ambient Room* illustrates how data can be integrated into the physical space in ways that respect the sensory, embodied, and atmospheric dimensions of human perception.

However, addressing the social nuances of cohabitation requires a departure from the *Ambient Room*'s original framing. The installation offered an important vision for information display, but its original framing remained grounded in an individual workspace context and focused more on informational awareness than on interpersonal mediation. This thesis repositions ambient light within the relational dynamics of student housing, where light does not communicate productivity-related cues but instead mediates ambiguous temporal states, interpersonal intentions, and shared rhythms of domestic life. The project explores forms of coexistence that ambient systems rarely address in the *Ambient Room*'s utilitarian context, shifting from automated information tracking to social interpretation.

To contrast these foundational experiments with contemporary smart-home paradigms, the Google Nest ecosystem (e.g., Yang et al., 2014) represents a dominant model of domestic ubiquitous computing (appendix 02). Nest devices demonstrate the viability of embedding sensor networks into the home to show presence, detect routines, and automate responses. From this infrastructure, this thesis extracts an important insight: that managing awareness, presence, and shared spatial rhythms within domestic contexts remains a significant social challenge for technology.

Yet, this thesis critically departs from the Google Nest philosophy. The ecosystem relies on explicit interfaces (as seen in figure 1), continuous monitoring, and predictive automation grounded in efficiency and optimization. Such approaches risk translating complex social practices into quantifiable patterns shaped more by automated systems than by human interpretation.

In contrast, this thesis project deliberately preserves ambiguity, interpretability, and interpersonal flexibility by employing ambient light as a low-resolution mediator, instead of a mechanism for behavioral prediction.



Figure 1. Google Nest Hub

Taken together, these perspectives reframe coexistence as something shaped through rhythms, interpretation, and awareness instead of explicit coordination alone. Existing ambient and smart home approaches reveal both the possibilities and limitations of embedding computation into domestic environments, particularly when optimization and monitoring overshadow or ignore social sensitivity. At the same time, these precedents demonstrate the value of subtle mediation.

They show that meaningful interactions arise from atmospheric cues that support interpretation and help inhabitants align without demanding attention, not from direct control. These cases also reveal how domestic technologies shape social expectations, routines, and subtle negotiations of space, even when presented as neutral tools. This thesis frames ambient interaction as a relational practice instead of a purely technical one.

This research project builds upon those qualities while deliberately moving away from screens, explicit tracking, rigid automation, and surveillance-oriented systems, combining insights from theory and existing design work to establish its own atmospheric design perspective.

3. METHOD

This project adopts a Research-through-Design approach, in which knowledge is generated through the iterative evaluation of design artefacts (Zimmerman et al., 2010). Research-through-Design positions itself as a method of inquiry where making, testing, and reflecting become central to understanding the problem. In shared households, social and spatial unease is embedded in everyday routines, often making it difficult to capture through observation or verbal reporting alone. RtD enables these aspects to be explored through intervention, allowing the study of how ambient changes influence perception and behavior.

To ground this inquiry, the research uses contextual inquiry methods, which focus on understanding situated practices and behaviors within their real-life context (Beyer & Holtzblatt, 1998). Through a set of semi-structured interviews - a qualitative method that combines guided interview questions with following open-ended discussion - the research investigates how inhabitants experience collision points where routines, temporalities, and needs overlap.

3.1 Project plan

The methodological plan of this project was structured as a progression from contextual inquiry to iterative design exploration, ensuring that the final intervention remains anchored in the experience of users. The process began with the identification and mapping of collision points within student housing, utilizing semi-structured interviews and contextual mapping to uncover the contextual and temporal frustrations of inhabitants. The main goal of this analysis was to move beyond superficial descriptions, focusing on how misaligned rhythms and routines lead to interference, defining the essential design guidelines needed to shape a non-verbal intervention for the domestic space.

The project employed bodystorming as a central generative method to bridge the gap between theory and physical intervention, allowing for the exploration of embodied interaction through the enactment of living scenarios. This approach provided a felt understanding of how physical boundaries affect the sense of privacy, enabling the testing of transitions before the implementation of digital and physical artefacts. This generative phase was crucial for defining how design objects can act as mediators of coexistence without imposing explicit forms of control.

The subsequent phase involved the development of prototypes through physical computing, focusing on the materialization of intent through ambient light. Wizard of Oz techniques were utilized to simulate intelligent responses, in order to maintain iterative flexibility and prioritize user experience over technical complexity. This process allowed for the testing of different availability modes and non-verbal signals in real time, observing how these changes influence the regulation of social boundaries.

The plan culminated in user enactment sessions, where participants interacted with the functional system in communal environments to validate the efficacy of diffuse communication. The evaluation focused on the interpretability of the cues and the impact on the inhabitants' sense of autonomy and privacy.

The main goal was to understand how subtle interactions embedded in the environment can influence the negotiation and support perception, while aiming to reduce socio-spatial frictions. Figure 2 shows us a visual of this thesis' project plan.

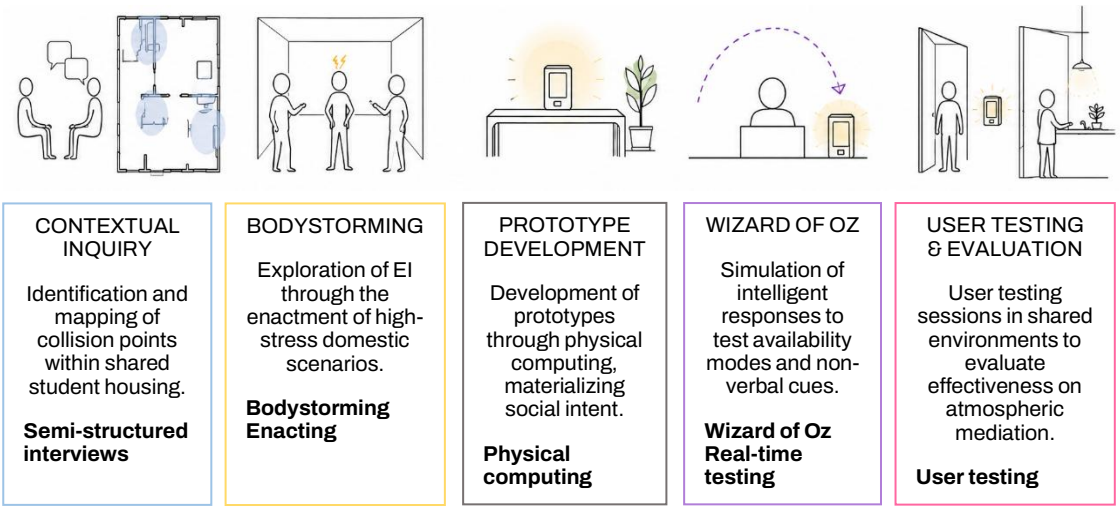


Figure 2. Visual of the project plan

3.2 Participants and context

Students were selected as a relevant group due to the transitional nature of their living conditions, where areas are often inhabited without established norms, requiring continuous adaptation due to the provisional situation.

In this framework, students are positioned as "extreme users" (Djajadiningrat et al., 2000) of domestic environments. Unlike family households, where cohabitation is often governed by established norms, student housing represents a more unstable and specific context. The transitional nature of these living arrangements means that norms are frequently undefined, intensifying the need for a constant interpersonal negotiation. This makes student housing a context where tensions become more visible and pronounced.

The study was conducted in Copenhagen, Denmark, within a broader European context of student housing conditions that are increasingly characterized by high-density cohabitation. Participants were international university students of different genders, all currently living in shared apartments of three to five residents. This sample reflects a typical student housing configuration in which temporary, mixed-background cohabitation is common, and where norms are still in formation. This allows the study to capture everyday relational dynamics and practices under structural constraints of student accommodation, despite differences in cultural and educational backgrounds.

3.3 Ethical considerations

Since this study involved participants engaging within private contexts to discuss interpersonal dynamics, ethical rules were applied throughout the process. Participation was entirely voluntary; all participants provided informed consent and were briefed on their right to withdraw at any time, with anonymity preserved through pseudonyms. In compliance with GDPR, all personal data - interviews audio and testing visual records - was kept to a strict minimum, stored securely, and will be permanently deleted following the thesis grading.

Because tracking systems can easily shift from awareness into surveillance, these ethical boundaries directly shaped the artifact itself: the design deliberately utilized ambient, low-resolution indicators to support coordination while preventing feelings of monitoring or overexposure.

4. DESIGN PROCESS

4.1 Contextual understanding: Semi-structured interviews

A set of semi-structured interviews was conducted along with seven individuals living in different inhabited households to ground the research in real-world experience, all of them aged 20 to 25.

This research topic opens up a lot of different nuances due to the simple fact of working with people and all of the social questions this raises. The approach then aligns with generative design research methods that aim to uncover lived situated experiences and implicit behaviors (Sanders & Stappers, 2012). Semi-structured interviews were chosen for their flexibility, allowing participants to describe their living situations while enabling the researcher to guide the conversation toward relevant themes. This format was particularly suitable for the topic, since many of the most relevant insights emerged through participants' descriptions of everyday situations.

Participants were selected based on their current experience of shared living, ensuring that they regularly engage with the types of situated and social dynamics central to the research. The interviews focused on exploring how inhabitants experience and manage moments of overlap within shared living situations. Participants were encouraged to describe moments of hesitation, everyday situations, and typical behavioral adjustments.

The analysis focused on how participants experienced moments of uncertainty within shared settings. The interview material was analyzed through an affinity-based process, allowing patterns to emerge from the data (Beyer & Holtzblatt, 1998). The material was

grouped according to recurring situations through an abductive sensemaking process (Kolko, 2010). This synthesis served as the foundation for the subsequent ideation phase, translating abstract social inconveniences into design insights.

The analysis was conducted digitally, using a whiteboarding tool to map and organize the interview data (figure 3). This allowed for a flexible and free organization process, where individual observations were visually grouped into thematic areas. To maintain a connection with the situated nature of the research, this digital synthesis was constantly cross-referenced with the original transcripts, ensuring that the emerging patterns remained grounded in lived experiences.

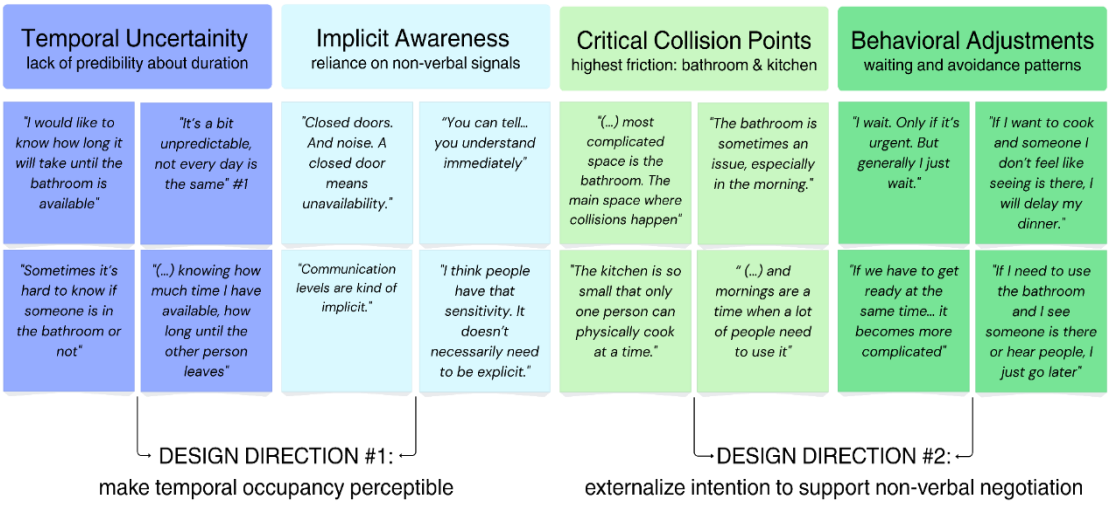


Figure 3. Affinity diagram: mapping recurring concerns regarding shared student housing

This process enabled the identification of underlying structures shaping how collective domestic space is used. Several recurring patterns were identified, revealing that these frictions are experienced as an ongoing process of anticipation and interpretation.

One of the most consistent findings is related to the role of temporal uncertainty and how it shapes everyday interactions. Participants did not describe sharing areas as problematic, but rather the inability to anticipate when a room would become available or how long it would remain occupied. This was particularly evident in the use of the bathroom, where activities are both time-sensitive and exclusive. As one participant noted, *"I would like to know how long it would take until the bathroom was available... it's a bit unpredictable, not every day is the same"* (Interviewee #2). This suggests that conflict emerges from uncertainty around temporal rhythms and availability, not from spatial limitation alone.

At the same time, participants consistently described relying on different forms of implicit awareness to navigate common environments. Signals such as closed doors,

noise, presence, or behavioral cues were used to interpret whether a place was available or whether interaction was appropriate. However, this awareness was often described as ambiguous, saying that it took too much effort and required continuous interpretation. As one participant stated, *"You can tell… you understand immediately"* (Interviewee #6), while another referred to *"closed doors and noise"* as indicators of availability (Interviewee #4). These findings highlight how collaboration depends on embodied and situated interaction (Dourish, 2001), yet remains fragile due to the lack of clear signals.

Across all interviews, the bathroom and the kitchen emerged as critical points of difficulty. Unlike other shared areas such as the living room, where overlap is easily tolerated, the bathroom operates as a setting of exclusivity, where only one person can be at a time. As one participant described, *"The most complicated space for me was the bathroom… that was where collisions happened"* (Interviewee #1).

Another recurring pattern concerns how participants manage these situations. Individuals tend to adapt their behavior through avoidance, delay, or passive waiting rather than engaging in explicit and verbal discussion. As one interviewee explained, *"I wait. Only if it's urgent. But generally, I just wait."* (Interviewee #1). This reflects a tendency to manage collective space through adjustment rather than direct communication, resonating with Altman's (1975) concept of privacy as a dynamic process of boundary regulation.

Taken together, these findings suggest that cohabitation is characterized by subtle forms of mediation drawn by uncertainty, some implicit communication, and temporal overlap. These findings point toward an opportunity for design interventions that enhance the perceptibility of time, availability, and state, supporting more fluid and peaceful forms of cohabitation.

4.2 From insights to design directions

Based on these findings, the project moved into a design exploration phase. To guide the exploration, the insights were translated into two design directions that investigate how ambient cues can support the daily navigation within shared settings. Each direction focuses on a specific dimension of the problem identified in the interviews, informing the development of multiple design concepts and prototypes.

#1 Make temporal occupancy perceptible. This direction focuses on designing traces that communicate not only if an interior is occupied, but for how long it is likely to remain so. The interviews show that the primary source of instability is not the access

to a determined location, but uncertainty about its duration. Making time perceptible supports anticipation and planning, reducing hesitation and unnecessary waiting.

This approach is grounded in temporal uncertainty as a core issue, as well as in patterns of passive waiting and the identification of the bathroom as a critical bottleneck. It aligns with Zerubavel's (1981) understanding that social coordination is structured through time, and Lefebvre's (2004) notion that friction emerges from collisions of rhythms.

The design intention here was to shift from binary states (occupied/free) to temporal perception (duration, flow, anticipation), allowing users to make decisions without interrupting others. This direction opens up a way for exploring how time can be represented and perceived within the environment. The main focus is on ambient cues that communicate duration and flow.

#2 Externalize intention to support non-verbal negotiation. This design direction focuses on designing subtle ways for users to externalize their intentions and needs in cohabited spaces, without requiring direct communication. While temporal uncertainty explains when disturbance occurs, a second layer emerges in how it is managed. Participants avoid direct alignment due to its social cost, relying instead on passive waiting and ambiguous signals. The issue is the absence of a clear, low-effort way to express their intent and not exactly a lack of sensitivity. This is reflected in statements such as *"You can tell… you understand immediately",* highlighting the reliance on implicit information.

The goal is to support ways of non-verbal regulation embedded in the environment, allowing users to externalize urgency, availability, or intent without direct interaction. This direction strengthens it by enabling lightweight forms of signaling through subtle changes in architectural elements or indicators of use. These concepts aim to reduce the social effort of the implicit nature of cohabitation, by making intention perceptible while preserving its low-friction and socially sensitive nature.

4.3 Iterative prototyping: Exploring cues and interactions

Guided by these directions, the project entered a prototyping phase to investigate how atmospheric cues can mediate temporal uncertainty and non-verbal negotiation. This stage involved developing exploratory concepts to understand how the ambient could materialize social intent. These ideas aimed to create social reactions and investigate how environmental changes influence the collisions of shared living. The process sought to explore the experiential qualities of these interactions in practice through three design concepts, by iterating across multiple approaches and adding nuances while enacting them.

Under design direction #1, the first exploration is an ambient light system placed outside of a room that gradually shifts in warmth and intensity as the room remains occupied, as seen in figure 4. The thought behind this intervention is to move away from the traditional binary "occupied/vacant" information. Instead, it tests if users can interpret duration through perceptual indicators.

By observing the gradual evolution of the light, an inhabitant can intuitively estimate if the person inside is "just finishing" or "just starting", allowing for better personal planning and reducing the social anxiety associated with unpredictable waiting.



Figure 4. Design concept 1.1: Temporal markers

For design direction #2, figures 5 and 6 show the two concepts that were developed to test the externalization of intention through non-intrusive and non-verbal mechanisms. Exploration 2.1 introduced a physical signaling system outside the bathroom. It allowed a waiting user to leave a persistent signal of their presence. This sought to inform the person inside that someone is waiting without the intrusive nature of knocking or verbal inquiry.



Figure 5. Design concept 2.1: Spatial transitions

Early design idea 2.2 shifts the agency to the person entering the space. By selecting a "mode", the user sets an ambient tone (within a scale) that communicates their expected duration to the outside (figure 6). This way, the person that's waiting can get a sense of the estimated duration of utilization, allowing them to manage their time and routine based on others' information. This tests the willingness to declare intent and whether such voluntary communication can solve socio-spatial friction.



Figure 6. Design idea 2.2: Diffuse presence indicator

4.4 Evaluative framework: Bodystorming and scenarios

The bodystorming and enacting sessions, involving students in an inhabited setting, provided a critical debate to explore the impact that the two parallel design directions had on domestic routines, acting as a driver for a deeper understanding of the boundaries between awareness and intrusion. These sessions aimed to simulate how the underdeveloped design ideas could work in real-life scenarios (figure 7), allowing the researcher to identify the strengths and weaknesses of each one of them.

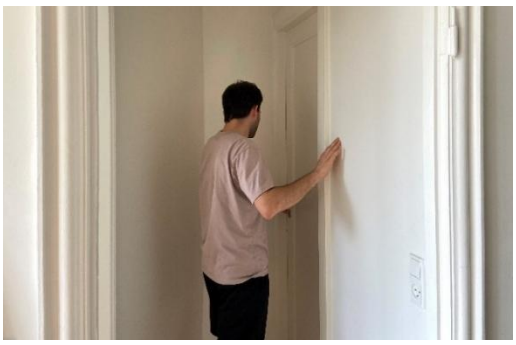


Figure 7. Bodystorming: simulating transitions in shared spaces

The evaluation of design exploration 1.1 reaffirmed the value of temporal predictability, as participants appreciated being able to intuitively estimate wait times through qualitative light progression. Some insights show that having a visual cue for the passage of time allowed them to plan their own routines with more autonomy, reducing the uncertainty typically associated with a closed door.

However, the enactment highlighted a sensitive point where a continuous and linear evolution of light could be perceived as a silent countdown, shifting the experience from ambient sensitivity to a form of surveillance that created unintended anxiety for the occupant.

Also, the lack of agency was described as a problem. Users felt uncomfortable with what the system might say about them, adding that it could become an involuntary sharing of habits, where awareness and privacy invasion might clash here.

The exploration of direction 2 provided insights into the regulation of agency and spatial management. Exploration 2.1 highlighted a significant social and organizational benefit: the ability to signal a need to enter without the inconvenience of a verbal interruption or a physical knock. Yet, the persistent and literal nature of the signal was sometimes felt as a direct and intrusive social pressure. While the main idea was for it to be subtle, participants felt it as a "ghostly presence" outside the door, creating an intrusive social pressure that broke their peace.

Similarly, 2.2 showed that while it was valuable to show intention through light, marking a specific point on a scale was too cognitively demanding for instinctive routines. Especially during high-pressure moments, such as the morning, the requirement to make a conscious decision about “how much time” one would take felt artificial and tiring, leading to a resistance toward the interface itself. Adding to that, they said that it felt “unnatural” to have this inside the bathroom and that they would rather have this outside, right by the light switch.

Participants who tested the concepts and enacted these scenarios highlighted the importance of agency and decision-making. A significant number of them described the positive sensation of having the power to signal intention.

Despite feeling a sense of invasion, all participants found the information useful for scheduling and organizing. For them, this double-sided experience represents a fair trade: sharing usage data in exchange for the opportunity to signal their own needs.

4.5 From iterative testing to integrated synthesis

Table 1: Comparative analysis of exploratory prototypes

Design	Strengths	Weaknesses	Key Takeaway
1.1	Temporal predictability; allowed for autonomous routine planning.	Perception of surveillance; "countdown" effect created anxiety on users.	Move away from "timer" logic toward spatial anchors.
2.1	Clear communication of wait-status without verbal interruption.	"Ghostly presence" felt as intrusive social pressure outside the door.	Transition to reciprocal (both-sided) atmospheric shifts and signaling.
2.2	Established the value of externalizing intent through light.	High cognitive load; explicit decision-making felt artificial and tiring.	Move towards intuitive gestures to capture intent.

These findings suggested that a truly responsive system should treat temporal awareness and social negotiation as a single experience and not as separate problems.

The enactment sessions indicated that the most effective mediation happens when information is felt rather than read. This realization led to the synthesis of the two directions into 3.0, a proposal designed to merge the positive attributes of each iteration while softening their limitations and redefining them.

In this version, the physical interface was refined into a minimalist, touch-sensitive surface where the occupant's intent is captured through the physical quality of their gesture upon entry. The system relies on an intuitive movement instead of a conscious selection: a quick, instinctive tap triggers a neutral light, signaling a brief occupancy. In contrast, a slower, elongated gesture activates a warmer amber tone. This allows the occupant to set the mood of the setting through a movement that mirrors their natural physical pace, removing the burden of explicit decision-making.

Outside, the system replaces the linear timer logic with an organic rhythm - a slow, breathing pulse of light. To balance it with the needs of those waiting, the prototype incorporates a reciprocal silent tap for the outside user. When activated, this signal causes a subtle ambient shift - a momentary oscillation in the internal light, instead of creating an alert like 2.1. This translates the social pressure of waiting into a gentle presence, allowing the occupant to recognize the other's need without a direct confrontation. Design idea 3.0 (seen on figure 8) moves away from the logic of notification toward a state of atmospheric attunement, where the environment itself acts as a mediator within the house. In this design artefact, the sense of agency and the double-sided (inside and outside) experience that users asked for were taken into account, allowing for the integration of the positive aspects highlighted in previous tests.

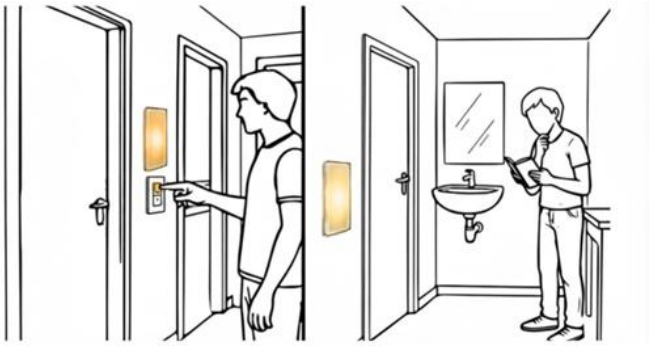


Figure 8. Design concept 3.0: Integrated ambient system

4.6 Wizard of Oz evaluation

Before implementing and developing technology for the final prototype, Wizard of Oz techniques were used to explore the potential benefits and failures from this project, simulating the behavior of design concept 3.0 in a controlled but realistic setting. The

aim was not to test technical performance, but to explore how participants would interpret and emotionally respond to the proposed mediation.

The simulation reproduced key interactions of the system, including internal light responses to entry gestures and an external silent tap mechanism, which was manually controlled by the researcher as we can see on figure 9. Participants were asked to engage with the system as if it were fully functional, allowing the study to focus on perception and behavioral adaptation.

A central topic that emerged during this stage was the question of temporal thresholds in signaling, particularly how long is “too long” when indicating intended occupancy of a common room. Participants raised this question when interacting with longer duration signals, often selecting the longer option when uncertain. Discussions suggested that acceptable duration is context-dependent and likely to be negotiated within each household over time, remaining flexible and not predefined.

One of the key explorations in this phase concerned the behavior of light as an atmospheric medium. An initial design idea for this system introduced a subtle pulsing light intended to simulate a sense of breathing occupancy. However, this approach did not perform as expected. Participants described the pulsing as visually distracting and cognitively demanding, particularly during waiting situations, where instead of creating calm perception, it introduced a sense of unnecessary activity. This informed the next iteration of the prototype, so the pulsing was removed and replaced with a stable single color light. This adjustment improved the subtlety of the system, suggesting that stability can communicate presence more effectively than continuous variation.

Participants responded to color more emotionally than as an exact indication of time. Warm tones were associated with slower, more private occupancy, while neutral tones suggested shorter or more functional use. Importantly, participants did not interpret color as a precise indicator of duration, but rather as a qualitative framing of the “type” of presence. Ambiguity here encouraged interpretation, with users constructing their own understanding of occupancy based on subtle changes in light behavior.

The external light also established a sense of background understanding between inside and outside users, allowing presence to be perceived without direct interruption. Once the pulsing was removed, this mechanism was described as subtle and non-intrusive, reinforcing a low-pressure awareness.

In addition, temporal parameters of the signaling system itself were evaluated in this Wizard of Oz. The duration of the external signal sent to the inside was tested across three states (1, 3 and 5 seconds). Participants consistently perceived 3 seconds as the

most balanced duration, with 1 second considered too brief to register and 5 seconds understood as slightly intrusive in the context of subtle communication.

The studies also addressed agency within the signaling system, specifically whether light signals should deactivate automatically or remain under user control. Participants would rather keep control over the deactivation process. Dismissing the signal manually felt more respectful and appropriate, suggesting that intention should be actively withdrawn rather than let it passively expire. This reinforced a key design principle of the concept 3.0: that agency and control in shared environments remain essential, even within ambient and non-verbal systems.

4.7 Material implementation of the prototype

Following the Wizard of Oz evaluation, the project moved from spatial simulation toward the material embodiment of the prototype. The goal of this phase was to translate the interaction into a physical artefact capable of mediating the relationship between inside and outside users through ambient cues. Insights gathered throughout the testing phase informed the transition from simulated interaction toward the development of a final integrated system, illustrated on figures 10 and 11:

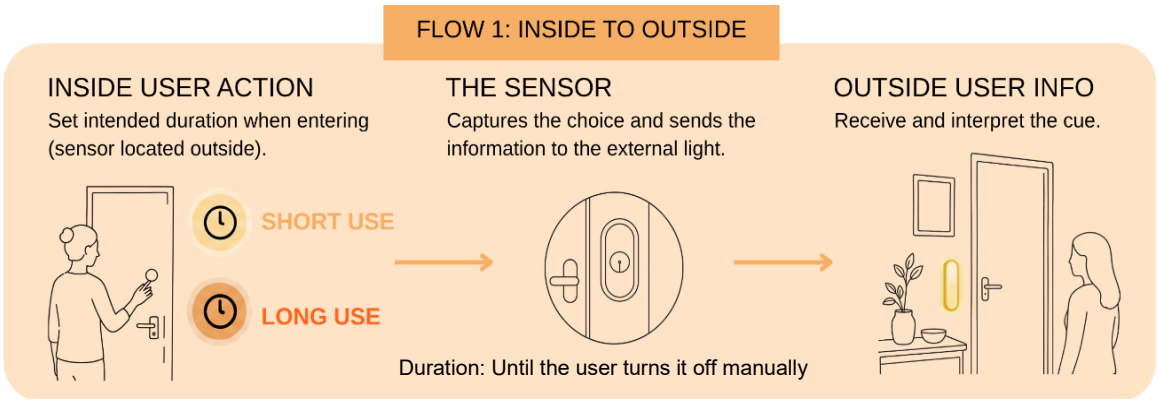


Figure 9. System diagram illustrating flow 1: inputs and outputs

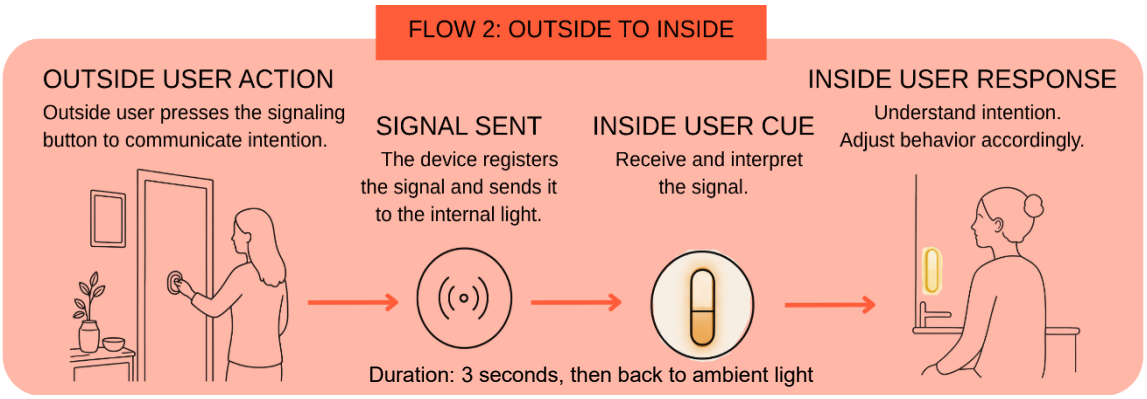


Figure 10. System diagram illustrating flow 2: inputs and outputs

This system was conceived as an embedded element integrated into the atmosphere itself, moving away from something that functions as a standalone notification. The intention was to subtly augment the environment with information through ambient light, instead of introducing another artefact demanding attention.

Within the project, light was seen as a design material capable of shaping atmosphere and social interpretation. Similar to how physical materials influence the experience of objects, the qualities of light - intensity, diffusion, and color - became central to how the interaction was felt and understood within the environment. The system aimed to explore warm and diffused lighting conditions capable of blending into the atmosphere without creating interruption or urgency, rather than using bright or saturated colors associated with alerts and notifications.

During Wizard of Oz evaluations and subsequent prototype testing, participants responded differently to variations in color intensity and diffusion, with stronger and more saturated tones often being perceived as intrusive or overly directive. Softer lighting conditions, in contrast, were more frequently interpreted as calm perceptual hints existing in the atmosphere.

From this perspective, light was approached as an ambient material capable of shaping the emotional and perceptual qualities of shared space (Zumthor, 2006).

The prototype was developed with the intention of being positioned close to the doors of collective locations, allowing its natural integration in the environment. Placed as a discreet device, it communicates through soft and unobtrusive ambient illumination.

The following sketch on figure 12 explores its physical embodiment within the environment adding to the relational logic of the interaction that the system diagram previously introduced. The intention was to create a discreet object capable of blending into the atmosphere of the home while still remaining perceptible through awareness.

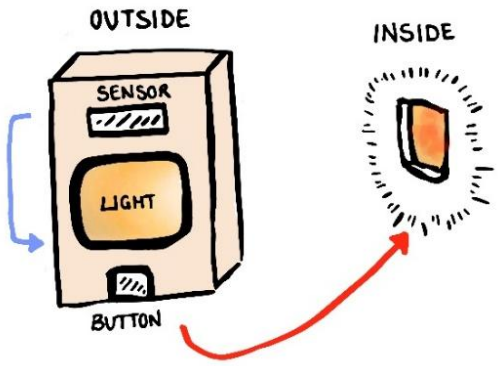


Figure 11. Sketch of final prototype: showing inside/outside interaction

The final prototype grows from users' insights and comments, merging enactments, tests, and reflections. It also emerges from participants' conclusions around the fair trade of understanding: the willingness to exchange small expressions of intention in order to gain a greater sense of predictability, autonomy, and coordination within the home. As a result, the final system integrates the project's two central design directions into a reciprocal inside/outside interaction model.

On one side, the external ambient light communicates occupancy and presence through soft cues (as we can see on the blue arrow), allowing the users to send a signal to the outside concerning the expected usage time, addressing Design Direction #1: *making temporal occupancy perceptible*.

On the other, the subtle signaling interaction allows intentions and urgency to be externalized non-verbally (orange arrow), with users being able to express their intention to use an occupied area by sending a non-intrusive light signal for 3 seconds, responding to Design Direction #2: s*upporting non-verbal negotiation*.

The prototype explores how small sensory interventions might soften the everyday tensions of shared living, supporting quieter and less cognitively demanding forms of coexistence. These interactions position the environment as a mediator of socio-spatial friction, transforming sites of uncertainty into spaces of surrounding awareness.

4.8 Technical development and integration

The chosen technical tool to bring this prototype to life was Arduino. The idea was to develop this double-sided system using sensors and actuators, such as LED lights, in order to achieve the calm technology that the whole project seeks to explore.

To have the two systems working at the same time, the plan was to use one button outside that would connect with an ambient light inside, signaling someone's intention to enter without actively disturbing them. At the same time, a sensor and a light outside would signal the intended time frame of the user inside.

This was tested and developed within an Arduino breadboard (figure 13), experimenting with different sensors and lighting configurations in order to understand how far the system could go while still maintaining the desired effect. Different sensors, such as LDRs, buttons, and motion sensors, were explored alongside variations in the number of LEDs, their positioning, intensity, and color. These experiments aimed to understand how subtle sensory information could communicate presence and intention without becoming intrusive or cognitively demanding.

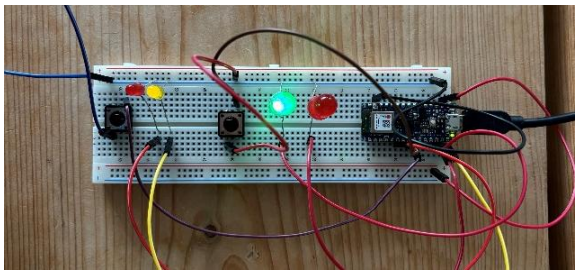


Figure 12. Running tests on Arduino breadboard: technical assembly

4.9 Iterations and refinements

Several adjustments were made during the testing of the final prototype to ensure the light transmitted a soft rather than directional effect, and that the device's physical scale integrated organically into domestic environments. This led to consecutive iterations on the final product, especially concerning its physical aspect. Details such as the light's size, shape, and enclosure thickness were refined along the way, printing different versions and sizes of the aimed final product. The inner geometry of the artefact was adjusted to increase the distance between the internal LEDs and the translucent front panel, in order to achieve a truly ambient effect. These physical iterations helped eliminate harsh light spots combined with subtle changes to the filament density, resulting in a uniform, cloud-like glow.

Additionally, the layout and performance of the touch-sensitive interface were refined to improve ergonomics and overall user experience. Different interface approaches and tactile exploration solutions were experimented on the outer surface, aiming to naturally facilitate the user’s movements during tapping gestures without the need for direct visual attention. These adjustments helped the object integrate more naturally into everyday use while supporting the system’s subtle interaction style. On figure 14 we can see a render of how it would look like in real life, with the device working as part of the environment. A sensor and a button on the outside signal the user’s intention, while the two lights respond accurately shifting very subtly towards the aimed state.



Figure 13. Conceptual rendering of the final design proposal

5. RESULTS AND DISCUSSION

User testing sessions were conducted after having the final physical version of the product (figure 15). In these sessions, participants interacted with the functional prototype in shared dwellings to validate the efficacy of atmospheric communication, revealing broader tensions around ambiguity beyond evaluating usability.



Figure 14. User testing the final physical prototype

The evaluation focused primarily on two core dimensions: the interpretability of the embedded cues and the impact of the system on the inhabitants' sense of autonomy and privacy. In response to the research question, the evaluations demonstrated that atmospheric interaction can mitigate socio-spatial friction by making occupancy and intention perceptible through ambient cues. The following sections unpack how subtle spatial and non-verbal interactions influence the negotiation of shared space, support peripheral consciousness, and ultimately reduce interpersonal tension.

5.1 Interpretation of atmospheric cues: Atmosphere vs. Precision

A central finding from the user enactments sessions was that the qualitative attributes of light functioned more effectively for coordination than explicit tracking systems. Participants demonstrated a strong preference for vague ambient light conditions over precise countdown indicators during the evaluation. Participants noted that earlier exploratory iterations (such as idea 1.1) that used a linear explicit progression of light to communicate duration introduced a rigid "countdown" effect. This interaction began to feel more like a monitoring system, generating pressure for the occupant to vacate the room.

The final prototype moved away from that by relying on stable ambient illumination, significantly improving the subtlety of communication. Participants understood color qualitatively, instead of interpreting the light as a precise measurement of time. The lighter yellow tone communicated a short use of the bathroom and therefore suggesting a shorter waiting period, while the warmer amber tone suggested a slower, more private occupation.

These results demonstrate how deliberate ambiguity can preserve the interpretative openness of the interaction. The system allowed occupants to perceive duration spatially by avoiding literal temporal representation. It supported softer forms of adaptation, allowing inhabitants to interpret situations contextually.

5.2 Impact on autonomy and privacy: Awareness vs. Surveillance

The deployment of the final prototype revealed a delicate tension between supporting awareness and unintentionally creating feelings of surveillance. A recurring insight throughout the project was that while social visibility can facilitate mediation, excessive perception easily transforms into social pressure. As participants repeatedly demonstrated during testing, *too much awareness becomes pressure; too much signaling becomes surveillance.*

The reciprocal interaction model of the design concept 3.0 helped mitigate this tension by reframing how waiting was communicated between inhabitants. Earlier iterations that explicitly exposed the presence of a person waiting outside the bathroom created uncomfortable forms of pressure for the occupant inside. In response, the final prototype translated the outside user's silent touch interaction into a brief and subtle oscillation of the internal ambient light. Participants described this ambient shift as gentle and non-intrusive. Presence was therefore felt rather than explicitly read, allowing the occupant to acknowledge the outside user without requiring direct verbal mediation or disruptive interruption.

Furthermore, the evaluation highlighted that keeping user agency was essential for the feelings of autonomy and privacy towards the system. Participants strongly preferred manually dismissing their selected availability states rather than allowing the system to deactivate automatically. Manual control was felt as more appropriate, reinforcing the idea that personal intentions should be actively communicated and withdrawn rather than passively managed through automation. Users associated manual dismissal with a sense of personal control over interaction.

In this sense, privacy emerged as a continuous process of regulating visibility within shared space, echoing Altman's (1975) understanding of boundary regulation.

5.3 Socio-spatial negotiation and coordination

The user enactments and testing of the final product demonstrated that embedding non-verbal interaction within the architecture of the home can meaningfully alter how cohabitation is negotiated. Prior contextual inquiry revealed that participants managed

domestic bottlenecks through passive strategies such as hesitation, avoidance, or waiting. Direct verbal negotiation was often avoided due to its perceived social cost.

The final prototype established what participants described as a “fair trade” of awareness. Inhabitants expressed a clear willingness to externalize small expressions of intent through quick and low-effort gestures because doing so provided a collective return in predictability and autonomy within the household. The interaction relied on embodied and intuitive actions, capturing intent through the natural pace of a tap rather than through explicit input. Consequently, the process of sharing information became trivial and embedded into daily activities.

This system enabled lightweight forms of coordination embedded within everyday routines. The ambient light set up outside served as a subtle mediator of socio-spatial understanding, making intention perceptible without calling for direct engagement. Instead of eliminating conflict through optimization, the intervention supported softer forms of adjustment that reduced uncertainty while preserving individual comfort and flexibility within shared space.

5.4 Design Implications

The empirical findings from this Research-through-Design process suggest several implications for designers developing atmospheric systems within high-density domestic environments.

First, the project indicates that ambient interaction may benefit from resisting highly precise forms of temporal representation and quantitative tracking. Participants consistently responded more positively to abstract and atmospheric expressions of duration than to explicit systems, which were often perceived as stressful or overly directive. The findings suggest that systems may benefit from relying on softer and more qualitative ambient transitions, such as gradual color blending and diffuse illumination, rather than communicating exact metrics. By maintaining a degree of ambiguity, the interaction leaves room for contextual interpretation, allowing inhabitants to adjust their behavior without feeling monitored or pressured by the system itself.

Building upon this, the evaluations also revealed the importance of reciprocity within atmospheric domestic interaction. Earlier interaction models that exposed waiting too explicitly often generated feelings of pressure for the occupant inside the bathroom. In contrast, interactions based on intentional gestures, such as touch, were perceived as more balanced and comfortable. The findings suggest that subtle forms of manual

participation may help preserve agency and autonomy within shared domestic environments.

Finally, the project highlights the importance of treating atmospheric systems as part of the environment. The physical placement, diffusion, and intensity of light strongly influenced how the interaction was perceived during testing. Softer and indirect illumination allowed the system to remain subtle within everyday routines, supporting peripheral awareness without demanding attention. These findings suggest that the physical integration of ambient systems plays an important role in how socially comfortable and non-intrusive they feel within domestic settings.

6. LIMITATIONS AND FUTURE WORK

The user testing and enactment dynamics provided highly valuable data regarding the efficacy of ambient mediation, but they also brought to light inherent limitations within both the system and the methodological framework itself. These constraints offer an essential perspective on the complexities of designing for shared contexts, while opening a clear path for future investigations in the field of calm technology and ambient interaction.

6.1 Limitations

The most critical limitation identified during the evaluation concerns the ethical nuances of privacy, transparency, and the potential risk of overexposure. Ambient interaction inherently operates on a delicate boundary between supporting peripheral awareness and inadvertently introducing a sense of surveillance. A deeper reflection on this topic reveals that privacy boundaries are deeply subjective and unstable, differing significantly from one individual to another. Consequently, the system faces the structural challenge of never being able to offer a single, static answer or a universal configuration regarding where each inhabitant's comfort limits lie. What one user interprets as a reassuring, non-intrusive sharing of presence, another might perceive as an uncomfortable exposure of their daily routines and personal rhythms.

Furthermore, the implementation demonstrated that the success of this system is intrinsically context-specific. The efficacy of this design does not rely solely on the functionality of the artefact, but rather on its integration into the pre-existing ecosystem of the household. The system depends heavily on the unique culture of the home and, above all, on the active and sustained willingness of the inhabitants to engage with it. If a resident consistently neglects to register their state or grows indifferent toward the ambient signals, the chain of mutual trust and transparency breaks down. Therefore, the device does not structurally resolve co-living conflicts on

its own; it merely mediates interactions that already require cooperation and social commitment from the cohabitants.

Additionally, interpretations and definitions of ambiguity, privacy, and communication are likely to vary across multiple cultural contexts, suggesting the need for eventual cross-cultural investigations into spatial interaction within communal dwellings.

6.2 Future Work

The principles explored in this thesis may also extend to other environments characterized by shared occupancy and temporal alignment, although situated within student housing. Future work should test the scalability of this model across different cohabitation typologies and living arrangements. Upcoming iterations could expand the intervention beyond bathroom transitions into more dynamic communal contexts, where task orchestration and resource sharing introduce different layers of interaction.

There is also room to explore how awareness diffuses across the wider domestic environment, allowing subtle occupancy signals to become perceptible from other rooms within the home through ambient cues. Beyond residential settings, the framework of atmospheric interaction may also extend into institutional and public contexts, such as campuses, coworking spaces, or offices.

In addition, later research should investigate how different demographics and age groups interpret ambiguity and adapt to non-verbal systems. There is also space to explore alternative atmospheric mediums beyond light, such as subtle variations in temperature.

Another important direction concerns the extent to which ambient interaction should become integrated into the architecture itself. While this thesis explored localized interventions, future work could investigate more immersive approaches where larger parts of the atmosphere become responsive to social dynamics. However, this also raises questions regarding saturation and overstimulation, particularly if cues become too pervasive within the home. Prospective studies should therefore examine the balance between ambient integration and domestic serenity when designing large-scale systems.

Finally, eventual studies should evaluate sustained use and the effects of habituation. Because the system relies on peripheral perception rather than explicit alerts, prolonged exposure may reduce users' sensitivity to the atmospheric cues over time. Observing how inhabitants engage with the system over longer periods would provide

further insight into evolving visibility boundaries, personalization, and long-term domestic interaction.

7. CONCLUSION

This thesis explored how atmospheric interaction design can support the negotiation of shared living through subtle and non-verbal forms of communication. The project investigated how moments of uncertainty around occupancy, availability, and waiting generate socio-spatial discomfort within routines, grounded in the context of student housing.

The research was guided by the question: *How can atmospheric interaction design mitigate social frictions in shared domestic environments?* This RtD process involved contextual inquiry, bodystorming, iterative prototyping, Wizard of Oz evaluations, and user enactments. Across these iterations, the project developed an ambient system that communicated presence and intention through light embedded in the home.

The project suggests that ambient interaction can reduce domestic friction by making social intention perceptible without making it overly explicit. The final prototype supported ways of collaboration through ambiguity and embodied interaction, rather than relying on quantified tracking or verbal interruption. Participants were able to interpret occupancy and waiting through atmospheric signals, allowing them to negotiate space with greater anticipation, flexibility, and autonomy.

A central contribution of the project lies in demonstrating that ambiguity can function as a productive design quality within residential interaction. Participants consistently responded more positively to qualitative cues than to precise representations of time or occupancy, which were frequently perceived as pressuring or intrusive. In this context, ambiguity preserved interpretative openness and reduced feelings of surveillance, allowing inhabitants to continue interpreting and adjusting situations socially, rather than having behavior rigidly defined.

Importantly, the project does not argue against interpretation itself. Despite the early stages of the research identifying uncertainty and implicit negotiation as major sources of discomfort, the evaluations revealed that completely removing ambiguity through explicit systems often introduced new forms of surveillance. Instead, the findings suggest that certain degrees of ambiguity can be socially beneficial when supported by meaningful atmospheric information. The system supported softer forms of interpretation rather than forcing inhabitants to interpret the unknown, grounded in subtle cues, allowing coordination to remain flexible and contextual.

The project also highlighted the importance of agency within ambient domestic systems. Participants preferred interactions that allowed them to actively regulate their own visibility and intention, reinforcing the idea that awareness must remain reciprocal and controllable in order to feel socially comfortable within living environments.

Ultimately, this thesis argues that the regulation of shared life does not necessarily require more automation or more behavioral optimization. Instead, it suggests that interaction design can support coexistence through quieter and more peripheral forms of environmental mediation. The project positions domestic technology not as a system of control, but as a subtle facilitator of everyday social coordination by integrating interaction into the domestic environment.

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To my mom, for remaining so close even 2500 km away. Thank you for being my rock since day one and for always making me feel missed and loved.

To my dad, for always encouraging me to dream as big as I want. Thank you for believing in me and for reminding me that every dream is worth pursuing.

To my grandparents, I hope I'm making you proud.

To my siblings, for being my greatest examples in life, and for inspiring me to become a little more like you every day. Your commitment and dedication have always given me something to look up to.

To my best friends, for making me miss home every single day. Thank you for giving me reasons to always come back.

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APPENDIX

01. Ambient Room: Ishii, H., Wisneski, C., Orbanes, J., Chun, B., & Paradiso, J. (1998). *Ambient Room*. MIT Media Lab.

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https://tangible.media.mit.edu/project/ambientroom/

02. Google Nest Hub: Google Nest Hub product demonstration video.

Video available at: https://www.youtube.com/watch?v=elXc31VDOlE

AI USAGE DISCLOSURE

Generative AI tools (Google Gemini, Microsoft Copilot) were used during the writing process to support language refinement, structural organization, and iterative feedback on argument clarity. All conceptual development, research decisions, analysis, design work, prototyping, and final interpretations remained the author's own responsibility. AI-generated suggestions were critically evaluated, edited, and adapted throughout the process.